

RevenueUsage Backend

Missing Features & Endpoints

Gap analysis comparing the RUTS desktop application against the RevenueUsage API backend.

RevenueUsage Backend — Missing Features & Endpoints

Purpose

RUTS is the existing WPF desktop application used by CBOS staff today. It has **no REST API of its own** — it talks directly to SQL Server via Entity Framework Core repositories and a Unit of Work (`Repositories\Unitofwork.cs`, `Repositories\RepositoryBase.cs` in the **RUTS** repo). Those repositories, together with the ViewModels/Views that call them, define the **full set of business capabilities** currently in production use.

RevenuUsage (this repo) is the new ASP.NET Core Clean Architecture backend (Controllers → MediatR CQRS handlers → Dapper repositories → stored procedures) intended to expose this functionality over HTTP for a future Angular client.

This document compares the two side by side, feature area by feature area, to track what still needs to be built in **RevenuUsage** before it can fully replace RUTS's data layer.

How to read the tables: each row is one capability. *RUTS behavior* cites the source repository/class in the **RUTS** repo. *RevenueUsage status* cites the controller/endpoint (or lack thereof) in this repo. *Gap* summarizes what's missing. *Priority* is a rough suggestion (High = blocks core workflows or is a hard dependency of other endpoints; Medium = important but not blocking; Low = nice-to-have / export-only).

Endpoint inventory used for comparison: 7 controllers / ~35 actions in `RevenuUsage.API\Controllers` (`TransferController`, `ResourcesController`, `ObligationsController`, `CurrenciesController`, `BeneficiariesController`, `LookupsController`, `RevenueUsageController`).

1. Authentication & Users — ****Missing entirely****

Feature	What RUTS does	RevenueUsage coverage	Gap	Priority
Login	Authenticates by employee number + password against the <code>Users</code> table (<code>Authentication\AuthenticateUser.cs:23-27</code>); routes to Employee vs Admin shell based on <code>Role</code> (<code>ViewModel\LoginViewModel.cs:116-123</code>)	None — no <code>AuthController</code> , no login endpoint anywhere in <code>RevenuUsage.API</code>	No way for a client to authenticate at all	High

Feature	What RUTS does	RevenueUsage coverage	Gap	Priority
User CRUD	Create/edit/soft-delete employee/admin users (<code>Repositories\UserRepository.cs</code> : <code>Exists</code> L40, <code>Update</code> L58)	None	No user-management endpoints	High
Password reset (self & admin-initiated)	<code>UserRepository.ResetPassword</code> L46, <code>CheckPassword</code> L53; separate self-service vs admin views	None	No reset-password endpoint	Medium
Role-based authorization (Employee / Admin / Super Admin)	Enforced client-side by which shell/menu loads (<code>Layout.xaml.cs</code> , <code>AdminLayout.xaml.cs</code>)	None — <code>launchSettings.json</code> has <code>anonymousAuthentication: true</code> , no <code>[Authorize]</code> attributes anywhere in <code>RevenueUsage.API</code>	No auth/authorization middleware or role checks	High

Everything else below assumes this gap will eventually be closed, since every write endpoint in RevenueUsage currently accepts a free-text `CreatedBy/ModifiedBy/DeletedBy` string in the request body instead of deriving identity from an authenticated principal.

2. Correspondents (master data) — ****Missing entirely****

Feature	What RUTS does	RevenueUsage coverage	Gap	Priority
Correspondent CRUD	Create/edit correspondent banks with Name, Code, Country (<code>Repositories\CorrespondentRepository.cs</code> : <code>Update</code> L65, <code>Exists</code> L23)	None — no <code>CorrespondentsController</code>	Can't create/manage correspondents at all	High
Correspondent combo/lookup data	<code>GetComboData</code> L29, <code>GetCorrespondentsWithAccountsComboData</code> L39, <code>GetCorrespondentName</code> L57	None	No lookup endpoint for correspondents	High

Note: RevenueUsage's `TransferController`, `ResourcesController`, and `CurrenciesController` all reference a `CorrespondentAccountId` as an input parameter, but there is no endpoint anywhere to create, list, or manage the correspondents or accounts those IDs point to. This is the most foundational gap — it blocks meaningful use of nearly every existing endpoint from a fresh database.

3. Correspondent Accounts (master data) — ****Missing entirely****

Feature	What RUTS does	RevenueUsage coverage	Gap	Priority
Account CRUD	Create/edit accounts per correspondent + currency (<code>Repositories\AccountRepository.cs</code> : <code>GetAll</code> L27, <code>Update</code> L77)	None — no <code>AccountsController</code>	No way to create/manage correspondent accounts	High

Feature	What RUTS does	RevenueUsage coverage	Gap	Priority
Account existence / movement checks	Exists L67, HasMovements L72 (used to guard delete)	None	No equivalent validation endpoint	Medium
Account balance	Tracked on the Account entity, updated by transactions/covers/deals	Read indirectly via statements/valuation only	No direct "get account" or "get balance" endpoint	Medium

4. Beneficiaries (master data) — ****Read-only in RevenueUsage****

Feature	What RUTS does	RevenueUsage coverage	Gap	Priority
Create/edit beneficiary	Repositories\BeneficiariesRepository.cs: Update L36	None	No create/update endpoint	Medium
List/combo beneficiaries	GetComboData L22, GetBeneficiaryName L29	None	No list/lookup endpoint	Medium
Beneficiary exists check	Exists L42	None	—	Low
Beneficiary statement (transfers)	Implicit via transaction history filtered by beneficiary	BeneficiariesController.GetBeneficiaryStatement (GET api/Beneficiaries/{beneficiaryId}/transfers)	Covered (read-only)	—

5. Currencies (master data) — ****Partially covered****

Feature	What RUTS does	RevenueUsage coverage	Gap	Priority
Currency CRUD (code, EN/short names)	Repositories\CurrencyRepository.cs: Update L119, Exists L125	None — CurrenciesController has no create/update/delete for currency master records	No currency master-data CRUD	Medium
Currency list / combo / by-correspondent	GetAllCurrencies L25, GetComboData L50, GetAvailableCurrencies L69, GetComboDataByCorrespondent L89	None	No "list currencies" endpoint	Medium
Currency balances	—	GET api/Currencies/balances	Covered (and arguably ahead of RUTS)	—
Correspondent balances by currency	—	GET api/Currencies/correspondent-balances	Covered	—
Daily valuation (cash/gold in USD)	Reserve reports	GET api/Currencies/daily-valuation	Covered (read)	—

Feature	What RUTS does	RevenueUsage coverage	Gap	Priority
Exchange rates (get/add/delete)	Rates read from <code>ExchangeRates</code> table, populated externally, only consumed via reports (<code>Report.GetRateFor/GetRateByIdFor</code> L507-528)	<code>GET/POST api/Currencies/exchange-rates</code> , <code>DELETE api/Currencies/exchange-rates/{id}</code>	RevenueUsage is ahead of RUTS here — full CRUD exists where RUTS only reads	— (covered / exceeded)

6. Resource Items (master data / categories) — ****Missing entirely****

Feature	What RUTS does	RevenueUsage coverage	Gap	Priority
Resource item CRUD	Create/edit resource categories used to classify credit transactions (<code>Repositories\ResourceItemRepository.cs: Update</code> L43)	None	<code>ResourcesController</code> only manages resource *transactions* (<code>AddResourceToCorrespondentAccount</code> , <code>GetResourceStatement</code> , <code>DeleteResource</code>) against a <code>ResourceTypeId</code> — there's no endpoint to create/list/edit the resource type/category master list itself	Medium
Resource item list/combo	<code>GetComboData</code> L27, <code>GetResourceName</code> L35	None	No lookup endpoint for <code>ResourceTypeId</code> values	Medium

7. Transactions / Transfers — ****Mostly covered, some gaps****

Feature	What RUTS does	RevenueUsage coverage	Gap	Priority
Create transaction (Credit/Debit)	<code>CreateTransactionViewModel</code> ; <code>TransactionRepository.Update</code> L27	<code>POST api/Transfer (TransferController.CreateTransfer)</code>	Roughly parallel; RevenueUsage transfer model doesn't explicitly surface a Credit/Debit "Type" the way RUTS does (relies on <code>OperationTypeId/ResourceTypeId/UsageTypeId</code> instead) — verify these map 1:1 to RUTS's Credit/Debit semantics	Low (naming/parity check)
Confirm transaction	<code>SuperTransactionViewModel</code> ~L498-516 (updates account balances)	<code>PUT api/Transfer/ConfirmTransfer</code>	Covered	—
Reject transaction	Not an explicit RUTS concept (soft-delete instead)	<code>PUT api/Transfer/RejectTransfer</code>	RevenueUsage is ahead here	—
Delete/soft-delete transaction	Soft delete via <code>Deleted_at</code>	<code>DELETE api/Transfer/{transferId}</code>	Covered	—

Feature	What RUTS does	RevenueUsage coverage	Gap	Priority
List/filter transactions (pending vs completed)	<code>TransactionsViewModel</code> splits pending (<code>StatementDate == null</code>) vs completed lists, with paging/filtering	None — no <code>GET</code> list endpoint for transfers, only statement/report-style reads	No generic "list transfers with filters" endpoint (only per-account/per-currency statements exist)	Medium
Account statement	Implicit from transaction list per account	<code>GET api/Transfer/Statement/GetCorrespondentAccountStatement</code>	Covered	—
Currency statement	<code>Report.GetCurrenciesData</code>	<code>GET api/Transfer/Statement/GetCurrencyStatement</code>	Covered	—
Final bank position	<code>Report.GetForeignReserveReport</code> (approx.)	<code>GET api/Transfer/Statement/GetFinalBankPosition</code>	Covered	—

8. Covers / Replenishments — ****Missing entirely****

Feature	What RUTS does	RevenueUsage coverage	Gap	Priority
Create cover (transfer between two correspondent accounts)	<code>CreateCoverViewModel</code> : <code>UpdateAccount</code> L191, <code>AddCovering</code> L201, <code>ExecuteSaveCommand</code> L207; updates both account balances	None	No equivalent command/endpoint anywhere in <code>RevenueUsage</code>	High
List covers	<code>Repositories\CoveringRepositories.cs: GetAllCoverings</code> L26	None	No listing endpoint	Medium
Validate accounts for cover	<code>CheckAccounts</code> L106	None	—	Low
Delete cover (with re-validation)	<code>CheckToDelete</code> L130, <code>Update</code> L173	None	—	Medium

9. Dealing (FX transfer between accounts) — ****Missing entirely****

Feature	What RUTS does	RevenueUsage coverage	Gap	Priority
Create deal (FX transfer with exchange rate)	<code>Repositories\DealRepository.cs : GetAllDeals</code> L25, <code>CheckAccounts</code> L90, <code>Update</code> L179	None	No <code>DealingController</code> /equivalent commands	High
List/delete deals	<code>CheckToDelete</code> L115, <code>GetComboData</code> L157	None	—	Medium

10. Reserves (Gold / Cash / Deposits) — ****Read-only in RevenueUsage****

Feature	What RUTS does	RevenueUsage coverage	Gap	Priority
Enter reserve stats (Gold, CashInHand, Deposits)	<code>Reserve\CreateViewModel</code> → <code>ReserveRepository</code> (base CRUD Add)	None — <code>CurrenciesController.GetDailyValuation</code> only reads cash/gold figures	No POST/create endpoint to record new reserve entries	High
Latest reserve stats	<code>Dashboard.GetStats</code> L116-120	Partially via <code>GET api/Currencies/daily-valuation</code>	Read side roughly covered; write side missing	—

11. Obligations — ****Payments covered, creation missing****

Feature	What RUTS does	RevenueUsage coverage	Gap	Priority
Create obligation (Bank / Company / Other)	<code>CreateObligationViewModel</code> ; <code>Repositories\ObligationRepository.cs: GetRate</code> L18, <code>Update</code> L24	None — <code>ObligationsController</code> only has payment/statement endpoints	No "create obligation" endpoint; obligations must already exist to record a payment against them, and the obligations table isn't even defined in this codebase's schema (only referenced via stored procs)	High
Add installment/payment	<code>InstallmentViewModel.ExecuteSaveCommand</code> L33-69	<code>POST api/Obligations/payment</code>	Covered	—
Delete payment	Soft delete	<code>DELETE api/Obligations/payment/{obligationPaymentId}</code>	Covered	—
Obligation statement (details + payments)	<code>Report.GetBanksObligationData</code> L367, <code>GetOthersObligationData</code> L415, <code>GetCompaniesObligationData</code> L461	<code>GET api/Obligations/statement/{obligationId}</code>	Covered for a single obligation; no aggregate "list all obligations by type" endpoint	Medium

12. Dashboard Analytics — ****Missing entirely****

Feature	What RUTS does	RevenueUsage coverage	Gap	Priority
Monthly debit bar chart	<code>Repositories\Dashboard.cs: GetBarChartData</code> L33-72	None	No analytics endpoint	Medium
Credit-by-resource pie chart	<code>GetPieChartData</code> L80-115	None	—	Medium
Correspondents balance widget	<code>GetCorrespondentsBalance</code> L74-78	Roughly covered by <code>GET api/Currencies/correspondent-balances</code>	Overlaps with an existing endpoint — may not need a dedicated dashboard route	Low
Latest stats summary (Gold/Cash/Deposits)	<code>GetStats</code> L116-120	Roughly covered by <code>GET api/Currencies/daily-valuation</code>	Overlaps	Low

13. Reports & Export — ****Partially covered, no export****

Feature	What RUTS does	RevenueUsage coverage	Gap	Priority
Correspondents balance report (multi-currency)	<code>Report.GetCorrespondentCurrenciesTotal</code> L31-131	<code>GET api/Currencies/correspondent-balances</code> (approx.)	Roughly covered; verify column/field parity	Low
Foreign reserve report	<code>Report.GetForeignReserveReport</code> L189+	<code>GET api/Transfer/Statement/GetFinalBankPosition</code> , <code>GET api/Currencies/daily-valuation</code> (approx.)	Roughly covered; verify parity	Low
Resources report	<code>Report.GetResourcesReport</code> L307	<code>GET api/Resources/statement/{correspondentAccountId}</code> is per-account only	No aggregate cross-account resources report	Medium
Credit/Debit movement reports	<code>Report.GetCreditMovements</code> L318, <code>GetDebitMovements</code> L343	Approximated by account/currency statements	No dedicated movement-list report across accounts	Medium
Obligations report (Banks/Companies/Others breakdown)	<code>Report.GetBanksObligationData</code> L367, <code>GetOthersObligationData</code> L415, <code>GetCompaniesObligationData</code> L461	Only single-obligation statement exists	No aggregate obligations report by party type	Medium
PDF export	<code>View\Reports*.xaml.cs: ExportToPdf</code> (CorrespondentsBalance L122, ForeignReserve L57, ObligationsReport L160)	None	No export/download endpoints of any kind	Low
Excel export	Same views: <code>ExportToExcel</code> (L253, L239, L339)	None	—	Low

14. Already covered / no action needed

These areas already have parity (or better) in RevenueUsage and require no immediate work:

Feature	RUTS	RevenueUsage
Banks (remitting banks) CRUD	<code>Repositories\BanksRepository.cs</code> (GetComboData, Update, Exists)	Full CRUD: <code>GET/POST/PUT/DELETE api/Lookups/banks/{id}</code>
Companies CRUD	<code>Repositories\CompanyRepository.cs</code>	Full CRUD: <code>GET/POST/PUT/DELETE api/Lookups/companies/{id}</code>

Feature	RUTS	RevenueUsage
Countries	Not a distinct master-data screen in RUTS (Country is just a field on Correspondent)	Full CRUD: <code>GET/POST/PUT/DELETE api/Lookups/countries/{id}</code> — a RevenueUsage addition beyond RUTS's scope, not a gap
Exchange rates	Read-only, populated externally into the <code>ExchangeRates</code> table	Full CRUD: <code>GET/POST/DELETE api/Currencies/exchange-rates</code> — RevenueUsage exceeds RUTS here
Beneficiary statement (read)	Implicit via transaction filtering	<code>GET api/Beneficiaries/{beneficiaryId}/transfers</code>
Health check	N/A (desktop app)	<code>GET api/RevenueUsage/HealthCheck</code>

Summary by priority

High priority (blocking / foundational):

- Authentication & user management (Section 1)
- Correspondents master data (Section 2)
- Correspondent Accounts master data (Section 3)
- Covers / Replenishments (Section 8)
- Dealing / FX transfers (Section 9)
- Reserve entry (create Gold/Cash/Deposits) (Section 10)
- Obligation creation (Section 11)

Medium priority:

- Beneficiary create/update/list (Section 4)
- Currency master-data CRUD (Section 5)
- Resource item master data (Section 6)
- Transaction list/filter endpoint (Section 7)
- Covers listing/delete (Section 8)
- Dealing listing/delete (Section 9)
- Aggregate obligations report (Section 11, 13)
- Dashboard bar/pie chart data (Section 12)

- Resources & movement reports (Section 13)

Low priority:

- Naming/semantic parity checks (Credit/Debit typing)
- Dashboard widgets that already overlap existing endpoints
- PDF/Excel export endpoints

Note on unused code in RevenueUsage: `RevenuUsage.Application\Features\RevenueUsages` (`RecordRevenueUsage` command, `GetRevenueUsageById` query, backed by an in-memory repository) has no controller wiring it to HTTP and does not correspond to any RUTS feature — it appears to be leftover scaffolding from the initial CQRS setup and is out of scope for this comparison.